NANYANG
THE INNOVATIVE POLYTECHNIC

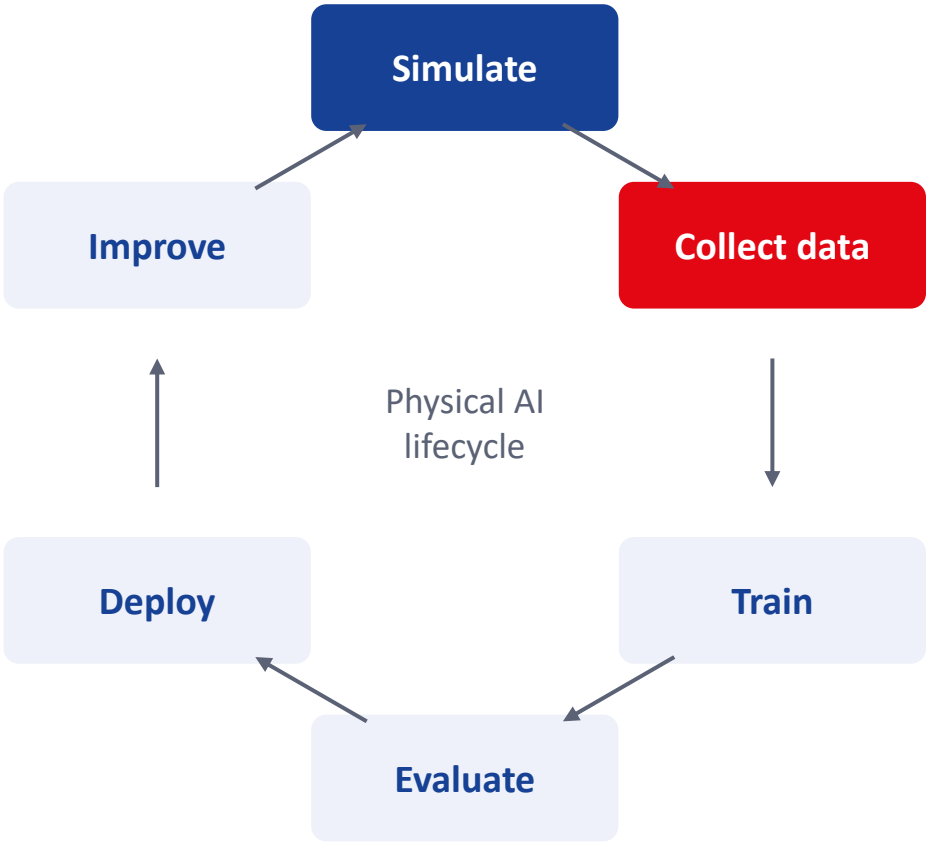
Robot Data

Teleoperation, kinesthetic teaching and simulation

Data Acquisition & Policy Learning · Lesson 3



Where we are: Block B · collecting robot data



A	Foundations	L1–L2
B	Collecting robot data	L3–L4
C	RL in simulation	L5–L6
D	RL: sim → real arm	L7–L8
E	Imitation learning	L9–L10
F	Evaluate & deploy IL	L11–L12
G	Vision-language-action	L13–L14

Entry check: three questions on Lab 2

1. What did calibration establish between the leader and the follower?
2. During teleoperation, what exactly does the follower copy from the leader?
3. At 30 Hz, how many joint values does the SO-101 report per second?

5 minutes, on paper or in the poll. We go through the answers together.

Learning outcomes

- By the end of this lesson you should be able to:
  - describe the structure of a robot dataset: episode, frame, observation, action, task
  - compare teleoperation, kinesthetic teaching and domain-randomised simulation on cost, realism and scale
  - apply quality and coverage rules to decide which demonstrations to keep
  - explain domain randomisation as a response to three kinds of reality gap

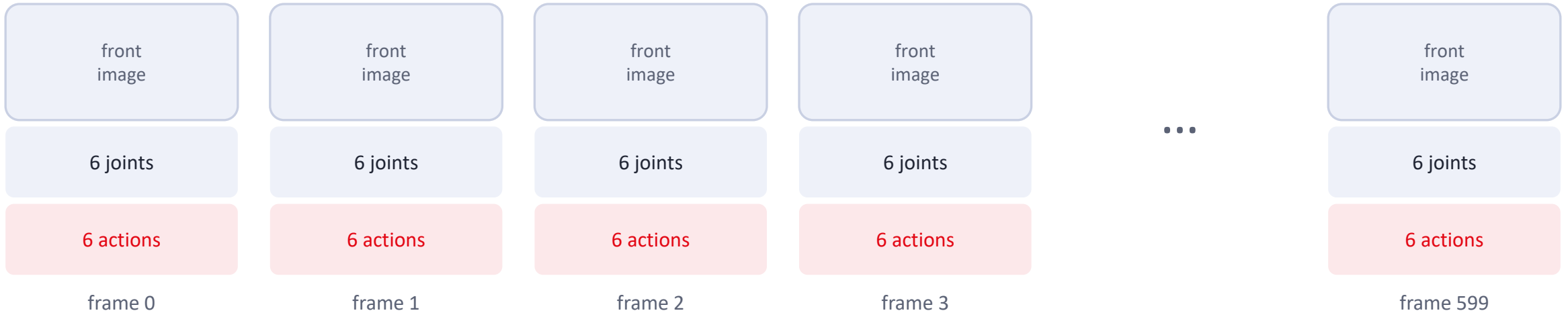
Part 1

What a robot dataset is

Episodes, frames, state and action

A robot dataset is a time series

One episode = one attempt at the task, from reset to finish (here 20 s × 30 fps = 600 frames)



Frame
one time step: observation + action + timestamp

Episode
an ordered sequence of frames for one attempt

Dataset
many episodes of the same task, plus metadata

Inside one frame

Field	Shape	Meaning
observation.state	6 numbers	follower joint positions
observation.images.front	480 × 640 × 3	front camera image
observation.images.wrist	480 × 640 × 3	wrist camera image
action	6 numbers	leader joint positions (the command)
timestamp · frame_index · episode_index	numbers	where this frame sits in time
task	text	“Pick the cube and place it in the bin”

Field names follow LeRobot's dataset format. Camera names (front, wrist) are chosen by you when recording and must stay the same.

State vs action: command and result



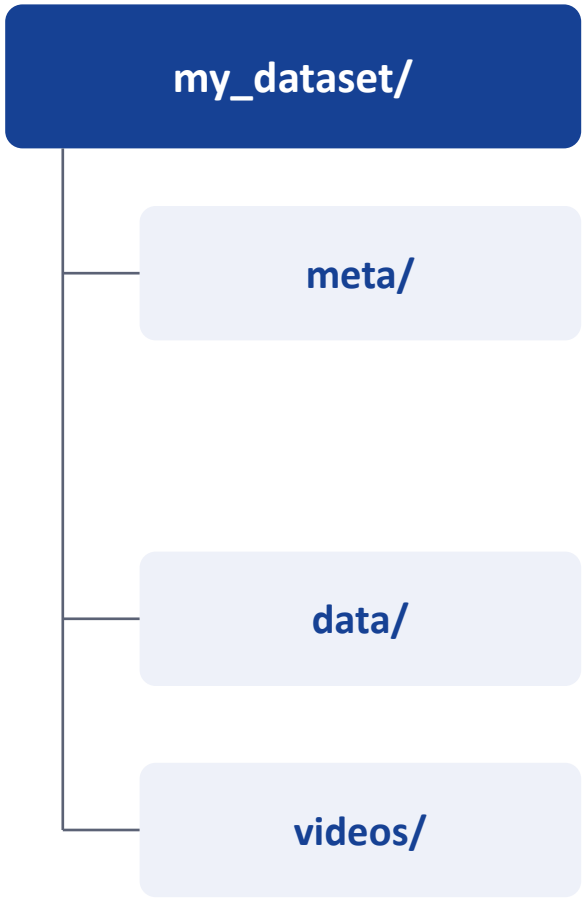
Illustrative curve. Replace with a joint trace from your own L2 teleoperation.

Action
where the follower was told to go

State
where the follower actually is

The follower lags a few frames and may not reach the command (for example when gripping). The policy learns to output commands.

How LeRobot stores a dataset (format v3.0)



info.json · schema, fps, features
stats.json · normalisation statistics
tasks · the task text
episodes/ · lengths and offsets

Parquet files · numbers for every frame (state, action, timestamps, indices)

MP4 files · one stream per camera (front, wrist)

Why this design?

Numbers are small and fast to read: Parquet.
Images are huge: stored as compressed video.
Many episodes per file: quick to load and stream from the Hugging Face Hub.

Normalisation: the same scaling at training and on the arm

During training
Each signal is rescaled using its mean and spread from stats.json, so the network sees numbers of similar size.

On the real arm
The same rescaling must be applied to live readings, and undone on the policy's output. LeRobot does this for you.

Example: elbow_flex mean = 60, spread = 20. A reading of 80 becomes (80 − 60) ÷ 20 = **+1.0**
If the scaling were different on the arm, the policy would see a different number for the same pose.

Worked example: how big is our dataset?

50 episodes × 20 seconds × 30 frames per second

30,000
frames

60,000
camera images (2 cameras)

180,000
joint values in the state

≈ 1 hour
to record, with resets

Compare: ImageNet has over a million images, collected by scraping. Every one of our 60,000 images comes from a person physically performing the task.

Part 2

Three ways to collect

Teleoperation, kinesthetic teaching and domain-randomised simulation

Three ways to collect robot data

Teleoperation

Control the robot through a separate device

- keyboard, gamepad, SpaceMouse, VR, or our SO-101 leader arm
- you never touch the robot that does the task
- the leader arm needs no retargeting: its joints match the follower's 1:1

Kinesthetic teaching

Physically guide the robot itself

- motors off or in a compliant mode; you push the arm through the task
- the robot records its own joint positions
- e.g. Franka guiding mode, UR Freedrive, SO-101 follower with torque off

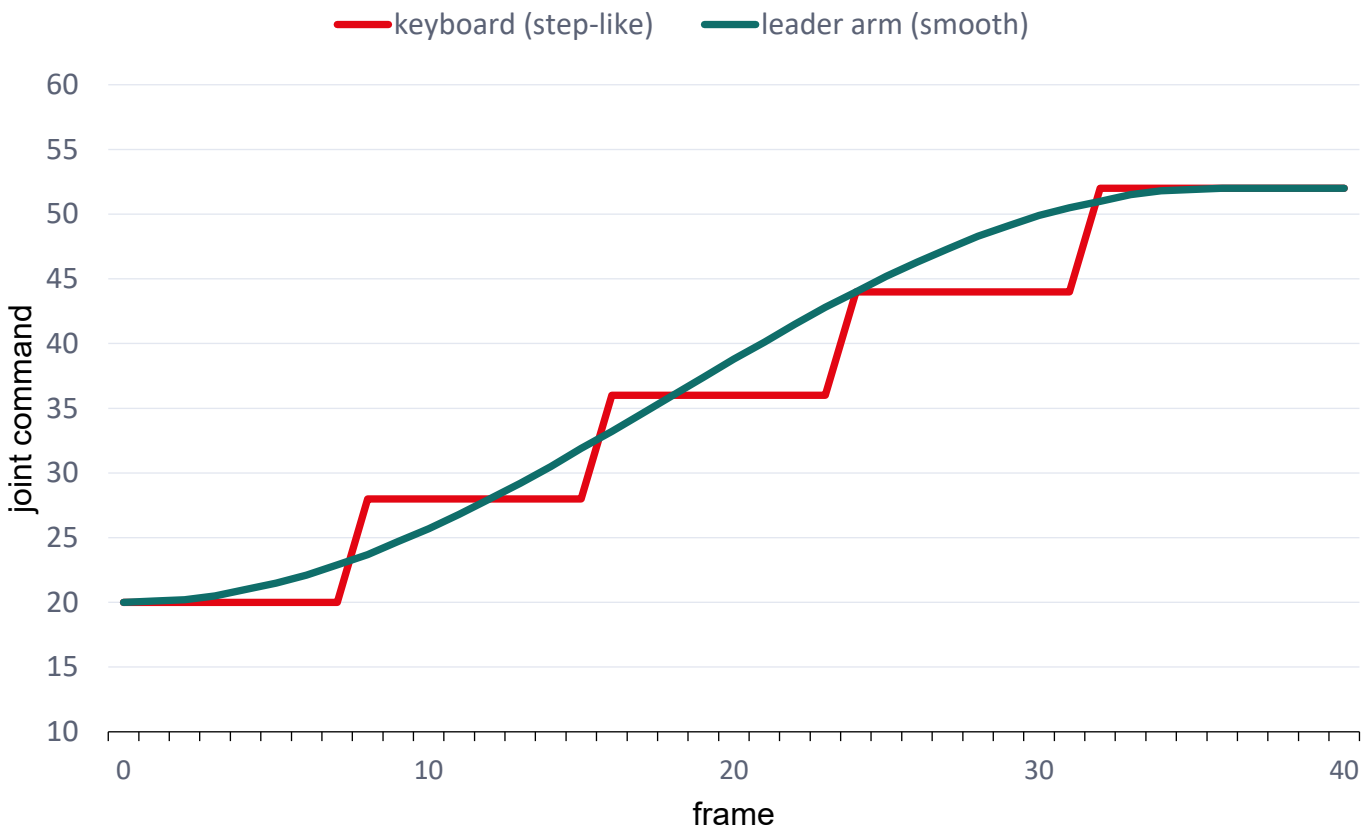
Domain-randomised simulation

Generate data in a varied virtual world

- physics, looks and sensor noise change every episode
- cheap, scalable, labels for free

The test: does your hand touch the robot that does the task? No → teleoperation. Yes → kinesthetic teaching. Leader–follower is teleoperation, as in LeRobot, ALOHA and GELLO.

Teleoperation: the device leaves a signature



Illustrative traces of the same reach: keyboard teleop vs leader-arm teleop.

What a device does to the data

- Keyboard: fixed-size steps, pauses between key presses
- Gamepad / SpaceMouse: smoother, but still retargeted
- VR hand tracking: natural, needs a headset and fast network
- SO-101 leader arm: smooth, joint-matched, no retargeting

The policy copies whatever style the data has.

Kinesthetic teaching: guiding the robot itself

PHOTO / VIDEO
Photo: a hand guiding the SO-101 follower with its torque off

Hand-guiding (lead-through)
torque off or compliant; push the robot itself and it records its joints

Record, then replay
replay the saved path with nobody in view to capture clean camera images

Strengths
direct feel of contact; no retargeting; no second device

Limits
your hand is in the camera images; tiring; you hold the arm's full weight

Domain-randomised simulation: many worlds, not one

PHOTO / VIDEO
Contact sheet: 16 randomised scenes from Isaac Lab

Physics: cube mass, table friction, motor stiffness

Appearance: colours, textures, lighting

Sensing: noise on joint readings, small offsets

Start state: where the cube is placed each episode

Idea from Tobin et al. (2017): make reality look like one more variation.

Comparing the three methods

	Teleop (incl. leader arm)	Kinesthetic (hand-guiding)	Randomised simulation
Cost per usable episode	medium (human time)	medium–high (human time)	very low once set up
Realism	real robot; style set by device	real robot; hand in camera view	simulated: reality gap
Scale	limited by people	limited by people	thousands in parallel
Who uses it in this course	Lab 4 (keyboard), L9–L14 (leader)	awareness; record-then-replay	RL training (L5–L8)

No method wins on everything. Real projects mix them.

Scaling up (for awareness)

Isaac Lab Mimic

Record about 10 human demos, split them into sub-tasks, and let the simulator re-stitch them in new scenes.

10 demos → 1,000 demos

LeIsaac

The real SO-101 leader arm drives a simulated SO-101, and the demos are recorded straight into LeRobot format.

leader-arm teleop + simulation

Mimic is mentioned only (Linux, GPU-heavy, not needed for our task). LeIsaac appears in Labs 4 and 10 where the lab machines support it.

Part 3

Quality, coverage and randomisation

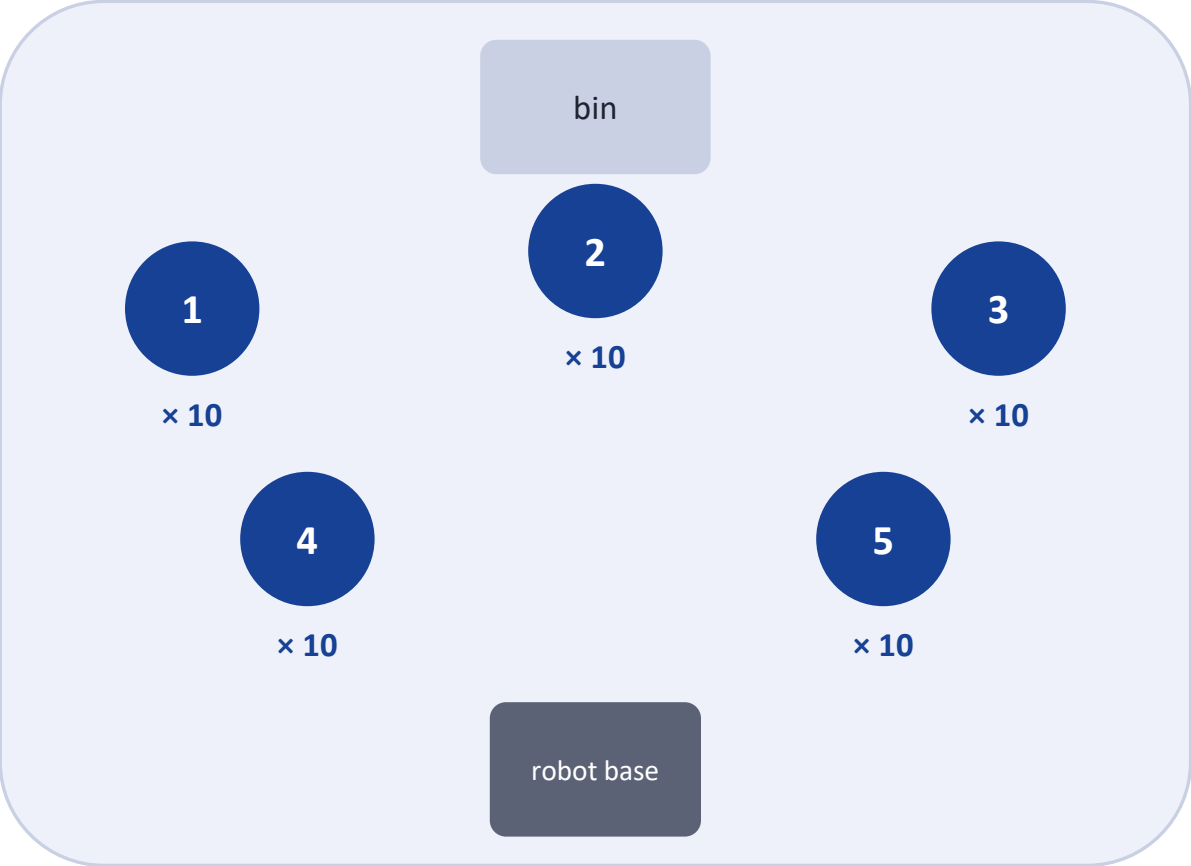
Deciding what to keep, and how wide to vary the world

Good demo, bad demo

Keep	Discard
• smooth and direct • one consistent grasp strategy • no pauses or hesitation • ends cleanly once the cube is in the bin	• pauses or long hesitation • regrasping or a dropped cube • a different strategy from the rest • cut off before the task finished

The policy copies everything, including the pauses and the regrasps. A bad demo is not neutral: it teaches the wrong behaviour.

Coverage: spread the demos over where the policy will work



Grid mat (taped down, never moved)

5 positions × 10 episodes = 50

25 episodes has been shown to be too few
(LeRobot's guidance for SO-100/101 tasks)

Fifty repetitions of one easy position teach the policy one position.

Curation: a written rule, not a feeling

Discard log (one row per episode)

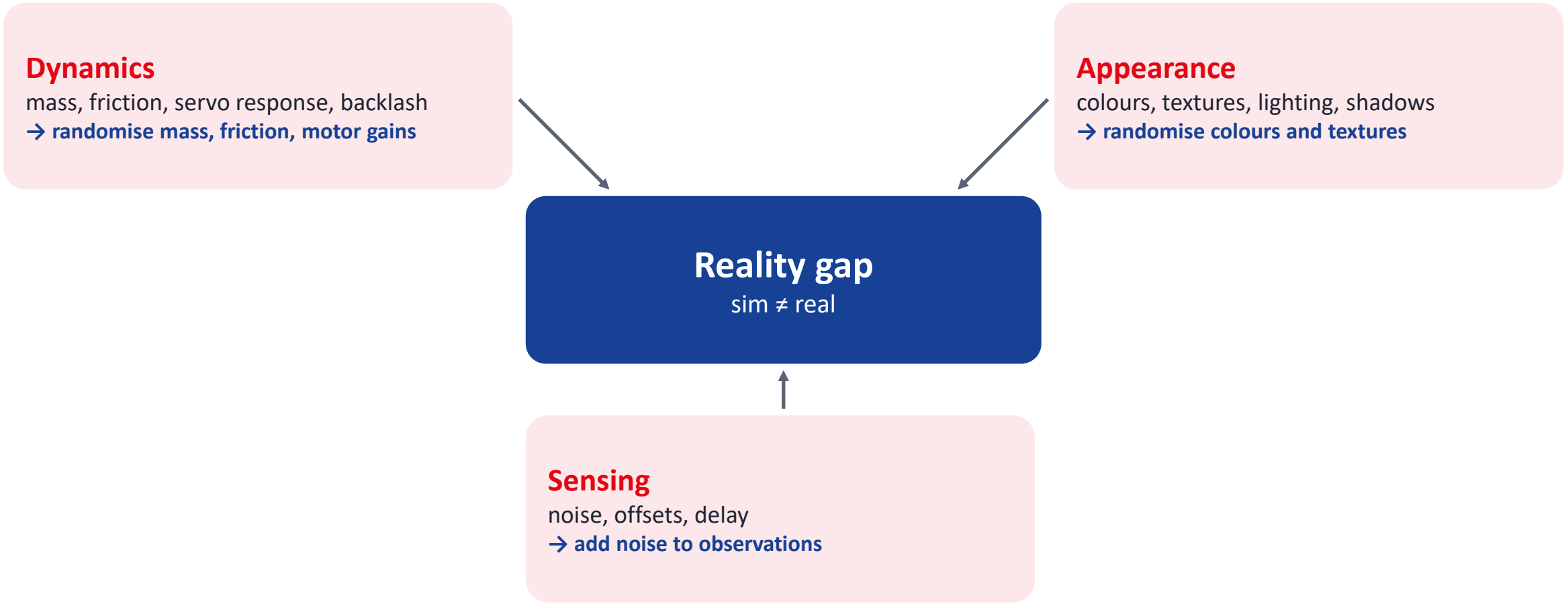
Ep	Keep?	Rule	Note
0	keep	–	
1	discard	pause > 1 s	operator distracted
2	keep	–	
3	discard	regrasp	cube slipped

Datasheet (ships with the dataset)

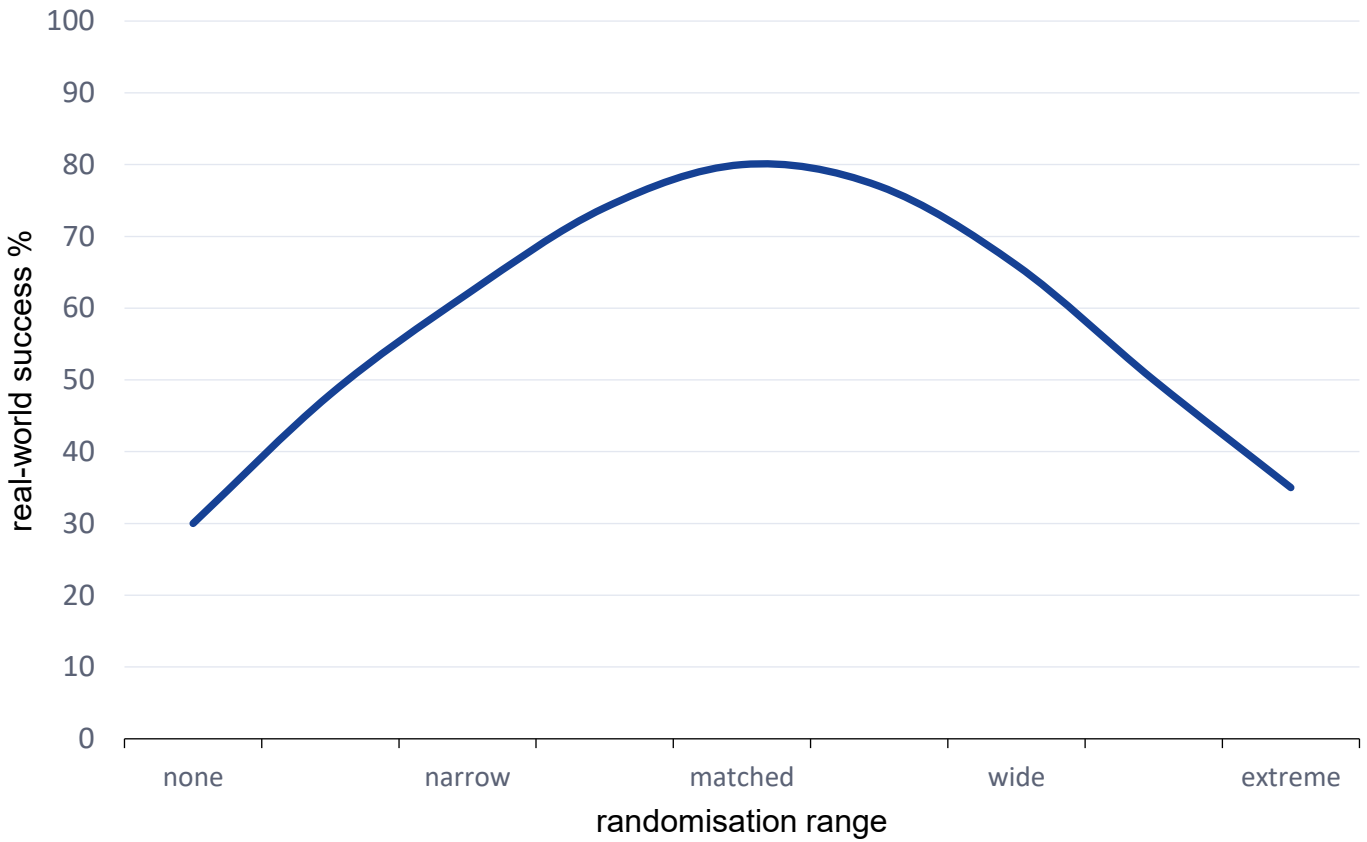
- task text and episode count
- coverage map (which positions, how many)
- camera set-up and names
- known defects
- the discard log

Based on “Datasheets for Datasets” (Gebru et al., 2018). Discarding is normal; hiding what you discarded is not.

The reality gap in three parts



How wide should the randomisation be?



Sketch of the trade-off; not measured data.

Too narrow
the policy overfits the simulator

Too wide
the task becomes hard to learn at all

Matched
cover what you cannot measure, sized to what you can (e.g. measured friction ± a margin)

Activity: Good demo, bad demo

Part A · six clips (10 min)

Watch each clip from the real arm.
Keep or discard? Name the rule.

Clip 1	Clip 2	Clip 3
Clip 4	Clip 5	Clip 6

Part B · choose a method (5 min)

1. Train a policy to reach targets using thousands of attempts.
2. Teach the arm a smooth pick-and-place from 50 examples.
3. Test whether a policy copes with different table colours before using the real arm.

Teleoperation, kinesthetic or randomised simulation? Why?

Exit ticket

- Answer before you leave:
  - A dataset has 40 episodes of 15 s at 30 fps with 2 cameras. How many frames and images?
  - Match each to the gap it closes: friction randomisation · colour randomisation · joint noise
  - State the coverage rule for our pick-and-place dataset.
  - What is the difference between state and action on the SO-101?

Next week: L4 lab · Collect data three ways

Sim station · Isaac Lab

Switch on randomisation; make a 16-scene contact sheet
Name the gap each term targets
Teleoperate the simulated arm and record 5 demos

Arm station · SO-101

Record 10 episodes with the leader arm
Replay one episode on the follower
Inspect the dataset and fill the quality log

Bring: your pod's calibration ids from Lab 2 (the same ids must be used for every recording).

References and further reading

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